BCC1244 - GAME PROGRAMMING



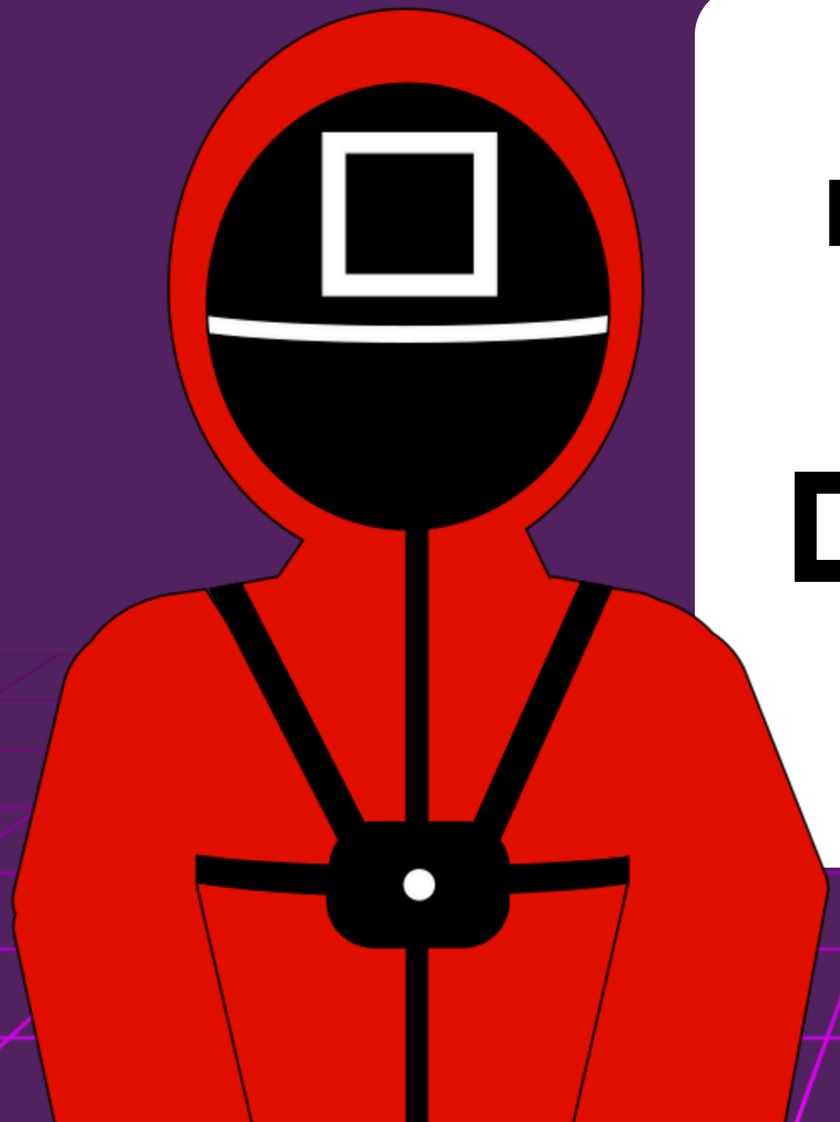


INTRODUCTION TO GAME PROGRAMMING

START









OVERVIEW OF THE GAME DEVELOPMENT PROCESS

OVERVIEW OF THE GAME DEVELOPMENT PROCESS



Game development is the process of creating a video game, encompassing everything from conceptualization to deployment.

Main stages of game development process:

1 - Concept and Ideation

2 - Pre-Production

3 - Production

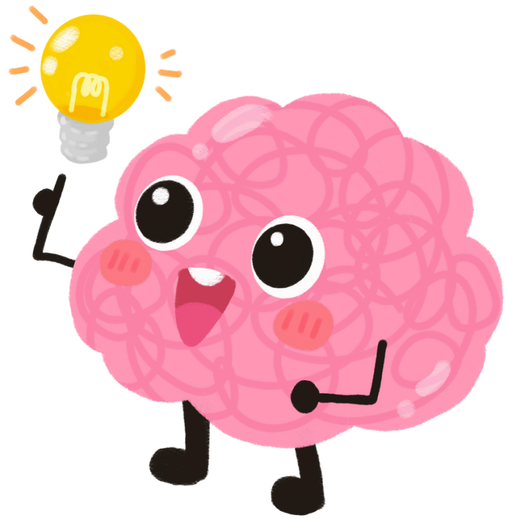
4 - Post-Production

OVERVIEW OF THE GAME DEVELOPMENT PROCESS

1 - Concept and Ideation

Idea Generation

Brainstorming unique and engaging game ideas.



Target Audience Identification

Determining who the game is for.

Concept Documentation

Writing a concise game concept document covering key features, gameplay mechanics, and themes.

OVERVIEW OF THE GAME DEVELOPMENT PROCESS

2 - Pre-Production

Game Design Document (GDD)

Detailed blueprint for the game, including mechanics, storyline, characters, art style, and technical requirements.

Prototyping

Creating simple versions of game mechanics to test feasibility and fun.

Team Formation

Assembling developers, artists, designers, and producers.



Overview of the Game Development Process

3 - Production

Asset Creation

Developing game assets, such as characters, environments, and sound effects.

Programming

Writing the code for gameplay, AI, physics, UI, and other functionalities.



Integration

Combining assets and code into a playable build.

Testing

Conducting Quality Assurance (QA) to identify and fix bugs and ensure a smooth player experience.

OVERVIEW OF THE GAME DEVELOPMENT PROCESS

4 - Post-Production

Polishing

Refining visuals, gameplay, and audio for a polished final product.

Marketing and Promotion

Building hype through trailers, social media, and collaborations.

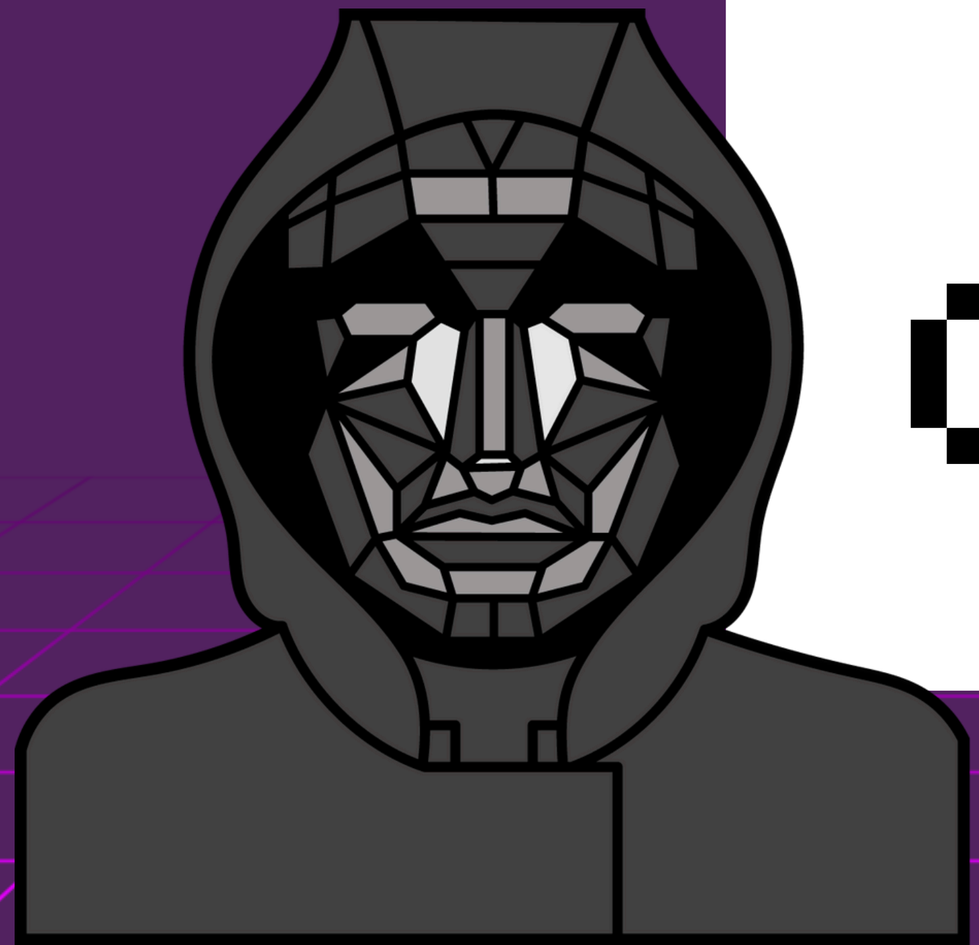
Release

Deploying the game to the target platform(s).

Maintenance

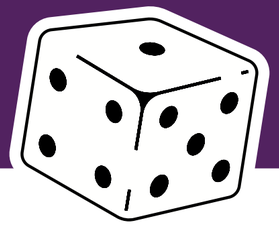
Providing updates, patches, and additional content post-launch.



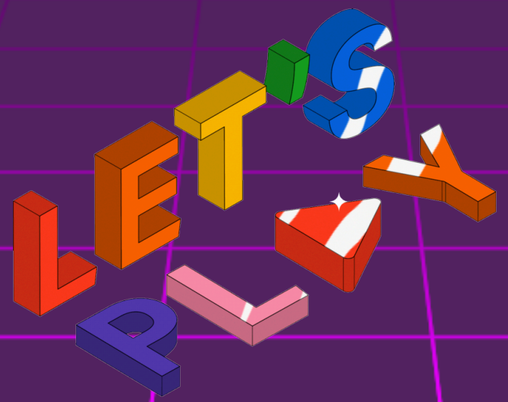


GAME GENAES

GAME GENRES



Game genres categorize games based on their gameplay mechanics, themes, and objectives. Understanding genres helps in designing games that appeal to specific player types.



GAME GENRES

ACTION

Subgenres: Platformers, Shooters, Fighting Games.

Key Features: Fast-paced gameplay, reflex-based challenges.



GAME GENRES

ADVENTURE

Subgenres: Narrative-driven, Puzzle Adventures.

Key Features: Exploration, storytelling, problem-solving.



GAME GENRES

ROLE PLAYING GAME (RPG)

Subgenres: Action RPGs, Turn-Based RPGs, MMORPGs.

Key Features: Character progression, immersive worlds, quest systems.

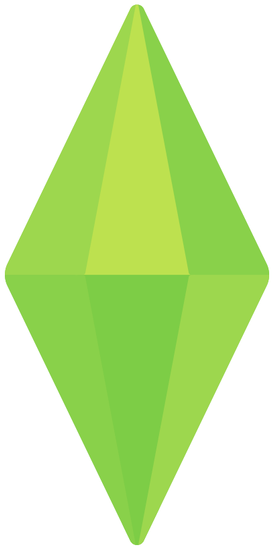


GAME GENRES

SIMULATION

Subgenres: Life Sims, Management Sims.

Key Features: Realistic gameplay focused on simulating real-world activities.

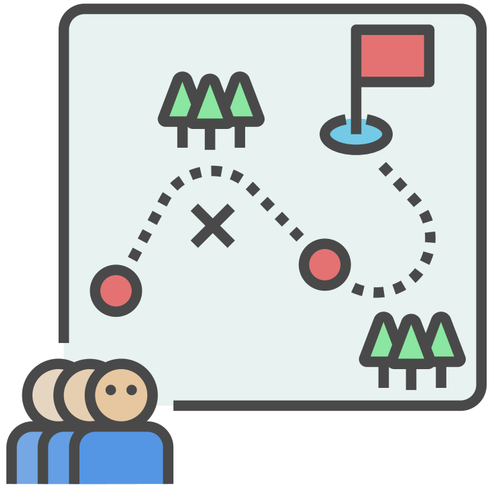


GAME GENRES

STRATEGY

Subgenres: Real-Time Strategy (RTS), Turn-Based Strategy (TBS).

Key Features: Resource management, tactical decision-making.



GAME GENRES

SPORTS & RACING

Subgenres: Team Sports, Racing Simulators, Arcade Sports.

Key Features: Replicating sports or racing experiences.

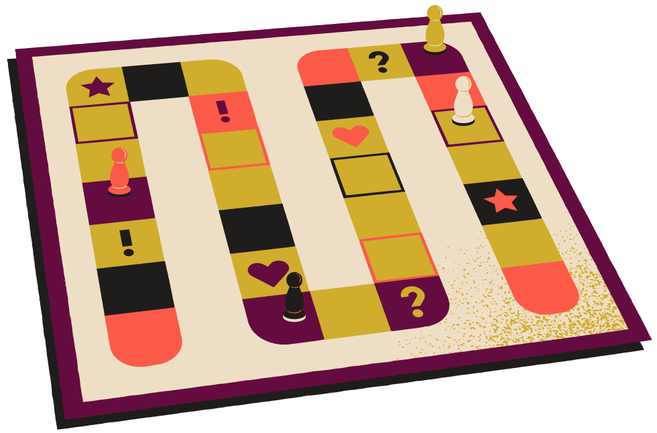


GAME GENRES

CASUAL & PARTY GAMES

Examples: Puzzle Games, Card Games, Trivia Games.

Key Features: Simple, accessible gameplay often designed for short sessions or group play.





GAME PLATFORMS

GAME PLATFORMS



The platform determines the hardware or software on which a game runs. Each platform has unique constraints and opportunities for developers.



GAME PLATFORMS

Personal Computer (PC)



Strengths:

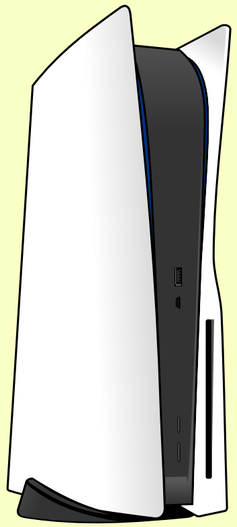
High performance, flexibility, modding support.

Challenges:

Diverse hardware configurations require optimization.

GAME PLATFORMS

Consoles



Strengths:

Standardized hardware, dedicated gaming ecosystem.

Challenges:

Licensing and platform-specific requirements.

GAME PLATFORMS

Mobile



Strengths:

Wide reach, accessibility, low barrier to entry.

Challenges:

Performance limitations, shorter play sessions, monetization.

GAME PLATFORMS

Web



Strengths:

Instant accessibility, no installation required.

Challenges:

Limited performance and graphics capabilities.

GAME PLATFORMS

VR (Virtual Reality) and AR (Augmented Reality)



Strengths:

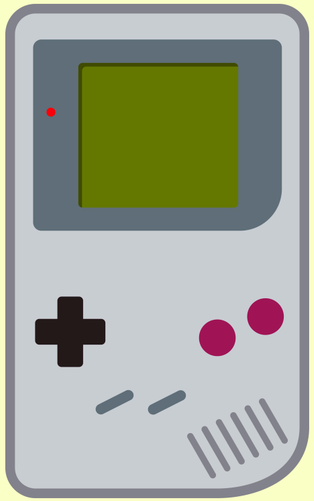
Immersive experiences, innovative gameplay.

Challenges:

High development costs, hardware adoption rates.

GAME PLATFORMS

Handheld Devices



Strengths:

Portability, unique control schemes.

Challenges:

Hardware limitations, niche audience.

GAME PLATFORMS

Arcade Machines



Strengths:

Social experiences, nostalgia factor.

Challenges:

Declining popularity, high development costs for custom hardware.

GAME PLATFORMS



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GAME PLATFORMS



SUMMARY





Understanding the game development process, genres, and platforms lays the foundation for creating engaging and marketable games. These elements guide decision-making throughout the development journey, ensuring the final product resonates with its intended audience and succeeds on the chosen platform.









THANK YOU FOR PLAYING!

